SUBARNA KARKI

Senior SDET | Test Automation Architecture | AI-Assisted Testing | CI/CD Quality Gates

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SUMMARY

Senior SDET with 8+ years of experience building test automation frameworks, developer tooling, and test data infrastructure. Sole QA engineer retained after Polly's 25+ person QA team was eliminated; owned quality end-to-end for ~50 developers and shifted testing to them with Claude Code tooling and self-healing test pipelines. Track record of measurable impact: release triage cut from 3 days to 1, ~2,500 test scenarios migrated from Cypress to Playwright in 4 months, nightly pass rate raised from 95% to 99.8%, and nightly regression time cut 85%. Specializes in test automation architecture, Playwright and Cypress framework design, and AI-assisted quality engineering.

SKILLS

Languages: TypeScript, JavaScript, Python, SQL, HTML5, CSS3

Test Automation: Playwright, Cypress, Selenium, Jest, Cucumber, Applitools, Page Object Model

Testing: E2E testing, API testing (REST, GraphQL), distributed systems testing, performance and load testing (Artillery), security testing, visual regression testing, test strategy and test plans, test data management, root-cause analysis, shift-left testing, Agile/Scrum

AI & Developer Tools: Claude Code (LLM), GitHub Copilot, Augment Code, MCP servers (Playwright, Jira, GitHub, Kibana), AI-assisted test generation, prompt engineering, self-healing tests, LLM-driven targeted test selection in CI

CI/CD: GitHub Actions, Jenkins, CircleCI, Harness; CI quality gates, parallel and sharded test execution

Cloud & Data: AWS (S3, EC2, Lambda, Secrets Manager), Docker, Kubernetes, MySQL, PostgreSQL, Grafana, SonarQube, Postman

Certifications: ISTQB CTFL, ITIL V3

Leadership & Process: Technical leadership, mentoring, coaching, quality metrics, release management, go/no-go release sign-off, quality roadmap planning, stakeholder management, test automation strategy, test infrastructure, hiring, budgeting, communication, collaboration

EXPERIENCE

Polly | San Francisco, California

Senior SDET | Sep 2024 - Nov 2026

Only QA engineer kept after the 25+ person QA team was let go in Feb 2026; sole quality owner for ~50 developers from then on.

- Cut release triage from 3 days to 1 day by building Claude Code skills that poll feature teams on automation status in the sprint release Slack channel and auto-heal broken tests, tracing each failure through GitHub commit and pull request history to its Jira ticket and the owning team, developer, or product owner, filing bugs and following them through to a verified fix. The skills also fetched and created Jira tickets, read and created Confluence pages, and pulled context from Slack; automated alerts assigned each failing test to its owning team via Playwright tags.
- Held go/no-go authority on release quality as the automation sign-off owner: releases waited until automation and manual blockers were fixed and retested.
- Owned the quality roadmap as sole QA engineer after the team elimination (Playwright migration, targeted PR test execution, AI-assisted test generation) and led the decision to migrate from Cypress with Cucumber to Playwright, delivering the migration in 4 months against a 6-month scope and moving ~2,500 test scenarios (an effort estimated at multiple years by hand). Owned both frameworks for ~2 years and ran both suites nightly in parallel during the migration to compare stability and performance before cutover; built the Page Object Model and fixture framework, Claude Code migration skills enforcing full assertion and test parity, and PR checks (lint, type-check, fixture coverage).

- Raised the nightly pass rate from 95% to 99.8% by fixing long-standing flaky tests and a pagination bug breaking suites across teams, and added pipelines that run tests on every pull request.
- Built an LLM-powered step in the CI pipeline that analyzes each pull request's code changes together with the related source files, then selects and runs the targeted Playwright tests for the affected areas, improving defect detection on changes before merge.
- Built a reusable Python test data (seed data) platform, now used by every engineer, that provisions fully configured mortgage org environments on demand across 30+ scenario types (pricing versions, lock policies, investor orgs, products, rules, channels), replacing manual setup that took a day or more per org. Fixed database locking that had caused timeouts when many data-creation scenarios ran at once, improving performance under parallel test load.
- Rolled out Claude Code to all ~50 developers - CLAUDE.md files covering framework and conventions, MCP servers for Playwright, Jira, GitHub, and Kibana, a skill that migrates hardcoded tests onto the seed data framework, and test-generation skills - achieving full adoption: every developer used it regularly and wrote automated tests for every pull request, so all shipped features had automation coverage, and AI-assisted test generation cut test authoring time from days to hours, sometimes minutes.
- Trained all engineering teams on Playwright through demos and hands-on training, and set the standards they follow: Page Object Model patterns, PR standards, test tagging, and a process keeping feature merges and automation merges in sync across separate repositories. Built a Claude Code code review skill, used by all engineers, that reviews test pull requests against those standards, backed by lint and CI checks enforcing proper fixtures, tags, and patterns.
- Cut tooling costs after the QA team elimination by auditing tools and licenses and cancelling what was no longer used, including X-Ray and BrowserStack.
- Built a test trends dashboard tracking pass rates, fail rates, and test trends by team for developers, managers, and release engineers, and reworked CI to merge sharded Playwright reports into a single report with shared caching.
- Wrote end-to-end Cypress automation for 6+ major Pricing Engine features (Counter Offers, MI Rate Cards, Lock Workflows, and more) that caught real bugs before release, plus a shared base class every Pricing Engine test now uses.

Infinite Blue (acquired by Everbridge) | Philadelphia, Pennsylvania

Senior SDET (promoted Apr 2024) | *Sep 2022 - Sep 2024*

- Cut nightly regression time 85% (2h 30m to 20m) by researching and rolling out Cypress Cloud and integrating it with Harness CI/CD for parallel execution across machines.
- Led the testing effort for a GraphQL security audit using standardized Postman collections; found and fixed critical issues (XSS, access control), including verifying that no API path allowed one organization to access another organization's data, and presented findings to leadership.
- Built and grew a Cypress framework for a React Single-SPA application covering 10+ micro front-ends and 10+ microservices, with custom commands, page objects, stable data-attribute locators, and tag-based pipeline runs for single-service testing. Added regression coverage immediately whenever a production issue was found, so the same failure could not recur silently.
- Built a Jest and Axios API test framework for the Cenari public REST API, using Node-Vault to generate API keys, org IDs, and test credentials, with scripts to seed and clean up test data.
- Ran performance and load tests with Artillery against public REST APIs under varied usage patterns. As the team's first SDET, ran the technical interviews that hired two SDETs onto separate teams and led their automation work as technical lead: sprint-by-sprint ticket reviews and automation planning, code reviews, pairing, and automation standards. Ran testing workshops for teams across the company, including Customer Support, and created a test plan template for all future projects, still in use.

Everfi | Washington, DC

Quality Engineer | *Mar 2021 - Sep 2022*

- Cut course development time from 3 days to half a day by building CLI automation tools and a shared NPM package of 20+ reusable Cypress functions for course components.
- Automated 13+ Everfi courses in Test Modeller, all running in CI, within the first quarter; wrote code and model organization guidelines; mentored new teammates, interns, and developers through code reviews and pairing; and served as the technical interviewer for new quality engineer candidates.

Gannett | USA TODAY Network | McLean, Virginia

Software Developer in Test | *Jan 2018 - Mar 2021*

- Shifted testing into development by teaching developers to use Simulato, a model-based tool that generates Selenium tests from models, reducing manual testing and catching bugs earlier.
- Added security scanning to GitHub Actions with SonarQube, flagging vulnerabilities, bugs, lint errors, and outdated dependencies, and added Selenium visual tests to Jenkins running in Docker containers as pipeline steps; created static fixtures to mock API calls for consistent visual and integration test data.

EDUCATION

Bachelor of Science in Software Development, Western Governors University

Associate of Applied Science in Information Systems Technology, Northern Virginia Community College